

Response variables: $\mathbf{Y} = (y_1, \dots, y_q)$

Predictor variables: $\mathbf{X} = (x_1, \dots, x_p)$

		Quantitative		Discrete	
		q=1	q>1	q=1	q>1
Quantitative	p=1	Simple regression	Multivariate regression	Simple logistic regression	
	p>1	Multiple regression	Multivariate regression Canonical corr. Partial corr.	Mult. logistic regression Discriminant analysis	Multivariate logistic regression
Discrete	p=1	t-test 1-way ANOVA	Hotelling T ² 1-way MANOVA	Simple χ^2	Loglinear models
	p>1	Factorial ANOVA	Factorial MANOVA	Logit models Loglinear models	Loglinear models